

Frontiers in Articular Cartilage Treatment

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Articular cartilage defects in the knee (Fig. 97.1) are common and often result in pain and dysfunction. Over the past several decades, research efforts have focused on better understanding how to diagnose and treat these lesions. Comprised predominantly of type II collagen, articular cartilage is relatively avascular, depending on diffusion to obtain nutrients and oxygen, making spontaneous healing of articular cartilage defects exceedingly difficult.^{1,2} Importantly not all defects are symptomatic, as many are simply incidental in nature, found on diagnostic imaging studies or during diagnostic arthroscopies being performed for other diagnoses, such as anterior cruciate ligament (ACL) tears. In fact, such defects are found in more than 60% of patients undergoing arthroscopy of the knee.³⁻⁵ One of the first challenges, therefore, is determining which lesions should be treated, and which can simply be managed with “benign neglect.”

For most patients, the combination of a thorough history, focused physical examination, imaging studies, and if necessary, a diagnostic arthroscopy, can determine if a given cartilage injury is symptomatic and warrants treatment. Once the diagnosis is made, a variety of treatment options are available. The optimal management for a given cartilage lesion varies based on lesion location (femoral condyle, tibial plateau, patella, trochlea), size, containment/stability, chronicity, and associated knee pathologies, including meniscus deficiency, ligament insufficiency, and/or malalignment. Bipolar “kissing” lesions (i.e., corresponding lesion on medial femoral condyle and medial tibial plateau, or on the patella and trochlea) are especially difficult to manage. In addition, factors unique to the patient, including age, activity level, expectations, body mass index, history of prior treatment, and ability to comply with rehabilitation, may influence the treatment decision. The goals of the patient are especially important to consider, as these may impact decision-making. For example, treatment for a high-level athlete hoping to return to the same (or better) level of play may differ from treatment for a weekend warrior who has already undergone multiple procedures and is hoping to manage activities of daily living without pain and swelling. Finally, surgeon- and facility-specific factors, including surgeon experience and the availability of treatments/products, will impact clinical decision-making. In most areas within the United States, for example, allograft tissue is readily available and can be considered as part of the treatment algorithm, whereas in many countries around the world, allografts are unavailable, and thus treatment for two very similar patients with two very similar defects can vary simply based on location.

Treatment for articular cartilage defects includes both non-operative and operative options. Nonoperative options include activity modification, physical therapy with a focus on quadriceps and core strengthening, cryotherapy, oral nonsteroidal anti-inflammatory medications, and a variety of injectable agents, including corticosteroids, hyaluronic acid, and more recently, biologics. Biologic therapies have recently emerged as a potential treatment for a wide variety of orthopedic pathologies, including articular cartilage lesions, and can be administered both in the outpatient clinic setting as well as in the operating room. Biologic therapies, including platelet-rich plasma (PRP) and mesenchymal stem cell injections such as bone marrow aspirate concentrate (BMAC), can be given either as isolated treatments, or combined with a surgical procedure. Surgical procedures for articular cartilage defects have historically been broken down into palliative (débridement, chondroplasty), reparative (marrow stimulation including microfracture), and restorative (autologous chondrocyte implantation, osteochondral autograft/allograft) procedures.⁶⁻²³ This chapter will focus on emerging surgical techniques for articular cartilage treatment. The advantages, disadvantages, outcomes, and complications associated with newer reparative techniques, including enhanced/augmented microfracture, as well as emerging reconstructive techniques, including matrix-associated autologous chondrocyte implantation, minced cartilage products, and off-the-shelf osteochondral allograft products, will be discussed in detail.²⁴⁻⁴⁹ It should be noted that several of these emerging technologies and products have only recently been introduced in the United States, with many are unavailable unless the patient is enrolled in a clinical trial. The majority of such products have been introduced and studied in Europe and/or Asia prior to becoming available in the United States, and even then, products must be considered “minimally manipulated” or intended for “homologous use” in order to bypass the US Food and Drug Administration approval process.

SURGICAL MANAGEMENT

For any patient undergoing surgical management of an articular cartilage defect, it is critical to (1) discuss goals/expectations and (2) treat any associated malalignment, meniscus insufficiency, and/or ligament insufficiency.⁵⁰ Patients must understand the unclear natural history of articular cartilage lesions and sometimes unpredictable nature of articular cartilage lesion treatment, especially with respect to returning to high-level athletics. In

Abstract

Focal chondral lesions of the knee, particularly those involving the weight-bearing surfaces of the medial or lateral femoral condyles as well as those in the patella and trochlea, often result in pain, effusions, mechanical symptoms, and ultimately dysfunction and disability. In many cases, surgical intervention is helpful in reducing pain, improving function, and restoring normal joint mechanics. “Traditional” techniques include chondroplasty, microfracture, autologous chondrocyte implantation, osteochondral autograft transfer, and osteochondral allograft transplantation. Emerging techniques including augmented microfracture, scaffold and matrix-associated constructs, minced cartilage transplantations, matrix-associated autologous chondrocyte implantation, and off-the-shelf osteochondral allograft transplantations have been described. Additional research on these novel techniques and comparison to more “conventional” techniques is warranted.

Keywords

cartilage lesion
cartilage restoration
focal chondral defect
stem cells
cartilage transplantation

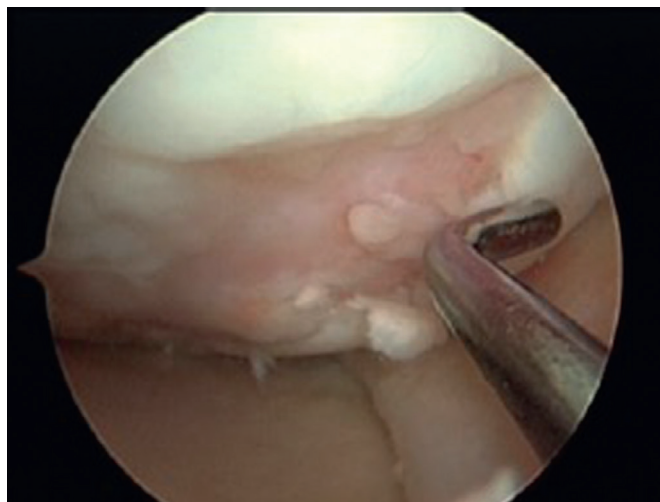


Fig. 97.1 Intraoperative example of a full thickness focal chondral defect of the medial femoral condyle (right knee).

addition, even with the most sophisticated articular cartilage treatments, unaddressed concomitant knee pathology is likely to result in failure of cartilage treatment, particularly in the setting of malalignment, as this may lead to overload of the newly treated cartilage lesion. While beyond the scope of this chapter, in some cases, isolated realignment osteotomy may be the ideal definitive solution for an articular cartilage lesion in a young patient with malalignment.

Palliative

Arthroscopic débridement, lavage, and chondroplasty continue to be viable treatment options for many patients with symptomatic, focal chondral defects who have failed nonoperative treatment. This approach is considered palliative as the goals are not to “regenerate” or replace cartilage, but rather to remove any flaps of loose cartilage that may be irritating to the patient and causing mechanical symptoms, and to stabilize the rim of the defect, decreasing the risk of further cartilage delamination. During the procedure, it is critical to débride the lesion down to the level of the subchondral bone, without violating the subchondral bone layer, taking care to maintain vertical walls around the lesion. Similar to a tire rolling over a pothole that is uniform and smooth along its rim, it follows that a well-performed chondroplasty will result in a defect with a uniform and smooth rim that is less likely to cause mechanical symptoms during loading and range of motion. Advantages of this technique include its technical simplicity, ability to be performed arthroscopically for defects in nearly any location (condyle, tibia, trochlea, patella), ability to perform other concomitant procedures such as meniscus surgery or ligament reconstruction, low overall cost, and ease of postoperative rehabilitation.⁷ The obvious disadvantage is that this procedure does not have any ability to replace the missing cartilage, and thus the patient may remain symptomatic. Chondroplasty has historically been considered an acceptable first-line treatment for low-demand patients with small articular cartilage lesions; however, this may also be appropriate for elite, high-demand patients who seek a minimally invasive approach with

a quick rehabilitation process and relatively early timeline for return to play.

Reparative

Another viable first-line treatment for patients with focal chondral defects is marrow stimulation. This approach is considered reparative, as the goals are to fill the cartilage defect with actual cartilage, as opposed to débriding the defect and leaving it “empty” as during chondroplasty. Marrow stimulation techniques include drilling/abrasion arthroplasty and microfracture (Fig. 97.2), widely considered the gold-standard surgical procedure for small, isolated focal chondral defects. As reported in a 2014 epidemiology study by McCormick and colleagues,⁵¹ surgical procedures for articular cartilage defects in the knee are increasing by approximately 5% on an annual basis in the United States, and of all coded procedures, microfracture remains the most common. The advantages of microfracture are similar to those of chondroplasty, including its technical simplicity, ability to be performed arthroscopically in a single-stage in a minimally invasive fashion, ability to perform other concomitant procedures such as meniscus surgery or ligament reconstruction, low overall cost, and different from chondroplasty, its ability to “fill” the defect with a cartilage product.⁵²⁻⁵⁷

From a biologic standpoint, microfracture and other marrow stimulation techniques induce an influx of marrow substrates to “fill” the cartilage defect, ultimately resulting in a fibrocartilage plug composed primarily of type I collagen. Importantly, fibrocartilage repair tissue lacks many of the intrinsic biochemical and viscoelastic properties of normal hyaline cartilage, is more stiff, and thus does not possess same shock absorption and force distribution capabilities as normal hyaline cartilage. When considering the pothole analogy described earlier, while the fibrocartilage produced by microfracture “fills” the defect, and may be superior to leaving the defect empty, the long-term efficacy of microfracture remains unclear due to the lack of hyaline-type cartilage (type II collagen) filling the void.

Recently efforts have been made to improve traditional microfracture techniques by using matrices and/or scaffolds to stabilize the mesenchymal clot produced by marrow stimulation, and to improve mesenchymal stem cell (MSC) differentiation into hyaline-type cartilage as opposed to fibrocartilage. Described “augmented microfracture”^{58,59} techniques include including BioCartilage (Arthrex, Inc., Naples, FL), autologous matrix-induced chondrogenesis (AMIC), BST-CarGel (Smith and Nephew Inc., Andover, MA), GelrinC (Regentis Biomaterials Ltd., Or-Akiva, Israel), and Chondrotissue (BioTissue AG, Zurich, Switzerland). The AMIC technique involves performing microfracture followed by the application of a porcine collagen I/III matrix (ChondroGide, Geistlich, Pharma AG) fixated with either autologous or allogeneic fibrin glue.⁶⁰⁻⁶⁴ The BST-CarGel technique involves performing traditional microfracture followed by the application of the product, which is a bioscaffold containing liquid chitosan and autologous whole blood.^{65,66} The GelrinC technique involves performing microfracture followed by the application of a hydrogel composed of polyethylene glycol di-acrylate (PEG-DA) and denatured fibrinogen, at which time the materials are exposed to UV light, forming a semisolid biodegradable scaffold for

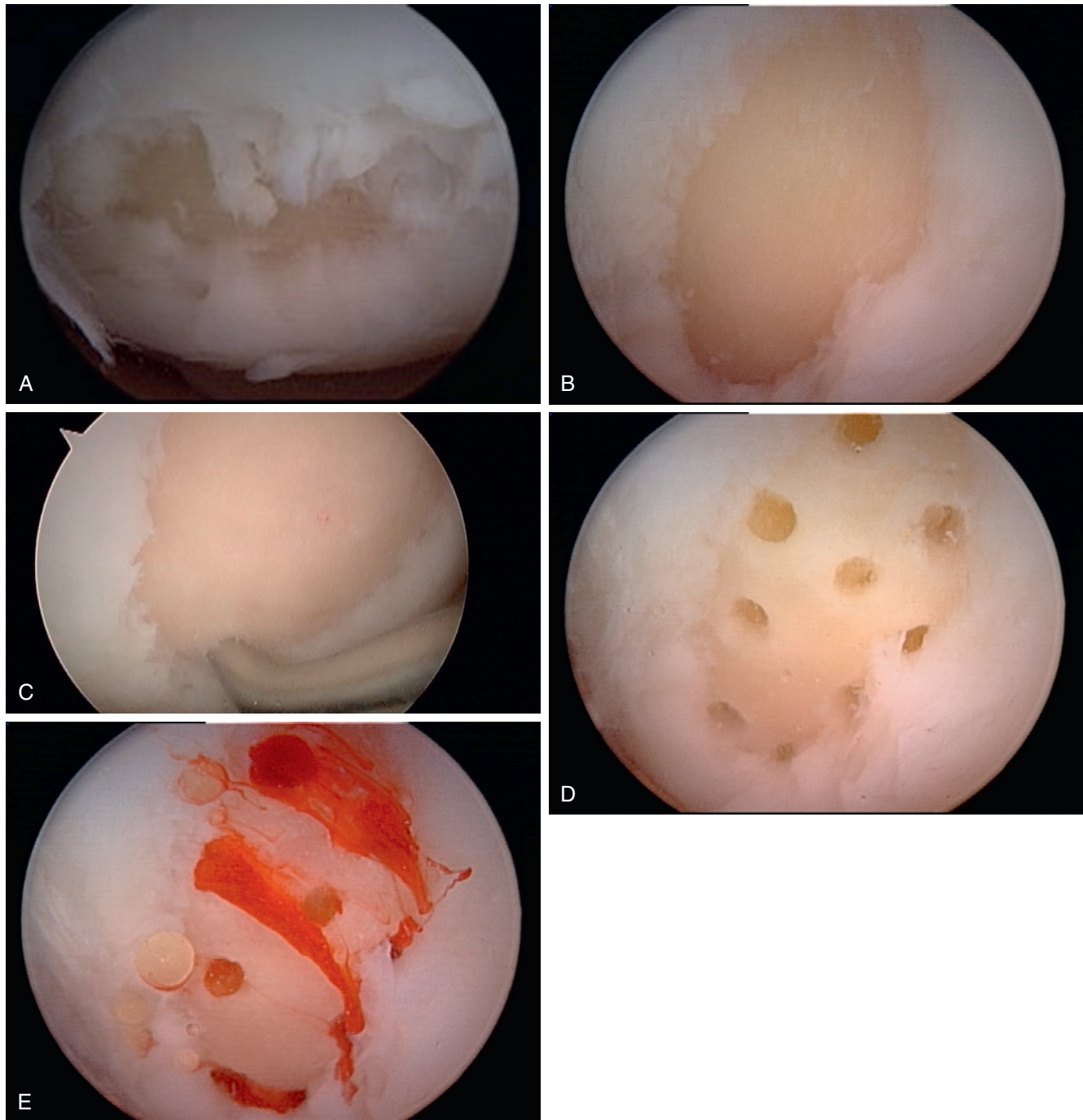


Fig. 97.2 Intraoperative example of a full thickness focal chondral defect of the lateral femoral condyle (right knee) undergoing microfracture including (A) defect, (B) débridement to stable vertical walls, (C) microfracture with an awl, (D) microfracture holes evenly spaced 3 to 4 mm apart, 3 to 4 mm deep, and (E) marrow products flowing into defect.

mesenchymal stem cells.⁶⁷ The Chondrotissue technique involves performing microfracture followed by application of a scaffold composed of polyglycolic acid (PGA) and hyaluronic acid (HA) immersed with PRP. Different from the aforementioned enhanced microfracture techniques/products, as of the time of publication of this text, BioCartilage (Fig. 97.3) is available for routine use in the United States. The BioCartilage technique involves performing microfracture followed by the application of 1 mL of

dehydrated, micronized allograft articular cartilage extracellular matrix combined with 1 mL of autologous PRP.^{48,49}

Each of these enhanced microfracture techniques are advantageous in that they are performed in the same general way as traditional microfracture, utilizing a single-stage, minimally invasive approach, and offer the theoretical benefit of improving the stability and biology of the defect repair site. These techniques can be performed either arthroscopically or through a mini-open

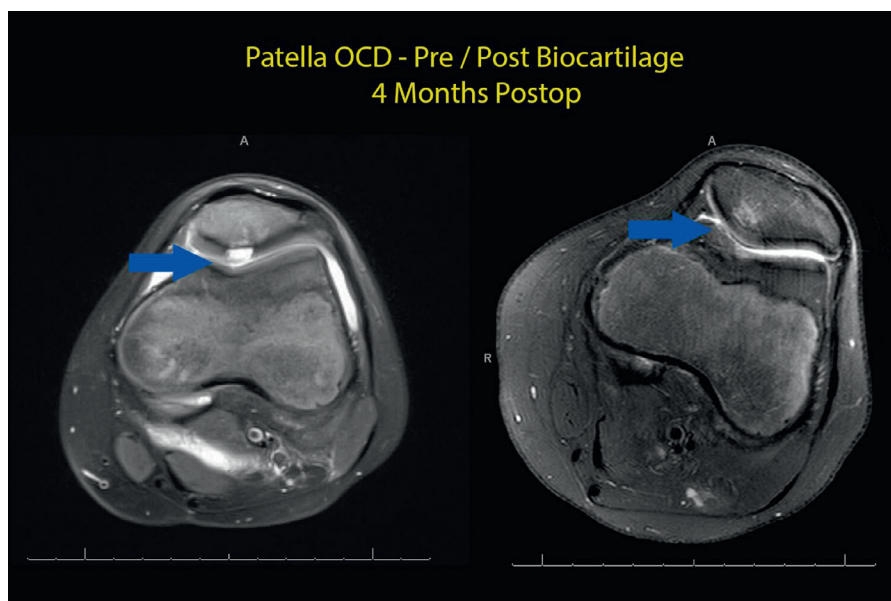


Fig. 97.3 MRI images pre- and post-BioCartilage procedure to the patella (4 months postoperative).

arthrotomy, depending on the ability to fully visualize and treat the lesion.

Restorative

For larger lesions, for lesions involving both the subchondral bone in addition to the cartilage, as well as for revision procedures, cartilage restoration surgery can be performed. A variety of restorative procedures are available, including autologous chondrocyte implantation (ACI), matrix-associated ACI (MACI), minced cartilage techniques, stem cell-scaffold techniques, osteochondral autograft transfer (OATS), surface allograft transplantation with “off-the-shelf” allograft products, and osteochondral allograft transplantation (OCA). These techniques are considered restorative, as they aim to treat the cartilage defect by restoring hyaline-type articular cartilage to entire defect site, either with or without subchondral bone. While many of the restorative techniques can be performed in a single-stage and through arthroscopic techniques, some require two separate procedures in a staged fashion, and some cannot be performed arthroscopically and require a small arthrotomy. The advantages of these procedures include their ability to treat large lesions as well as lesions that have failed prior attempts at cartilage repair, the presence of hyaline-type cartilage (as opposed fibrocartilage), and their ability to treat the subchondral bone in addition to the cartilage (for the osteochondral grafting procedures; Fig. 97.4). Disadvantages include the potential high cost associated with allograft tissue, the relatively long recovery period needed for graft incorporation, the potential safety concerns with allografts (for allograft procedures), and the need for two procedures instead of a single surgery in some cases (for ACI and MACI procedures). As ACI, MACI, minced techniques, OATS, and OCA were covered in the previous chapter, the following section of this chapter will focus on emerging articular cartilage techniques, including stem cell-scaffold techniques, novel minced techniques, and off-the-shelf surface allograft transplantation.

Mesenchymal Stem Cells With Three-Dimensional Matrices

Recently several unique products combining MSCs with three-dimensional scaffolds have been introduced in an effort to provide an alternative option to ACI for the management of focal chondral defects that, unlike ACI, can be performed in a single operation. Similar to ACI, the aim of these techniques is to treat the defect by restoring the surface with durable hyaline-type cartilage. Described products in this category include Hyalofast (Anika Therapeutics, Bedford, MA), which utilizes autologous MSCs, and Cartistem (Medipost Co., Ltd., Korea), which utilizes allogeneic MSCs. The thought process behind these techniques is that the scaffold will create an environment that is biologically favorable for MSC differentiation into hyaline-type cartilage.

Hyalofast, which is not currently available in the United States, is performed in a single stage by shaping/sizing the hyaluronan scaffold (HYAFF11 scaffold) to the defect shape/size, soaking the scaffold in the patient’s BMAC, and then securing the scaffold to the defect with 6-0 PDS suture and/or fibrin glue.⁶⁸ BMAC is most often harvested from the patient’s iliac crest, and contains adult MSCs, platelets, cytokines, bone morphogenic protein (BMP) 2 and 7, and a variety of growth factors, including PDGF and TGFβ. The cells, proteins, and growth factors are thought to establish a favorable biologic environment for cartilage restoration due to their anabolic and anti-inflammatory properties. While clinical data is limited, Gobbi and colleagues have found that at 5 years, patients undergoing treatment with Hyalofast compared with microfracture had better rates of returning to preinjury activity levels, despite microfracture patients having better return rates at 2 years following treatment.⁶⁸

Similar to Hyalofast, Cartistem is also performed in a single stage and combines MSCs with a three-dimensional scaffold. In contrast, Cartistem utilizes allogeneic stem cells, specifically culture-expanded human umbilical cord blood derived

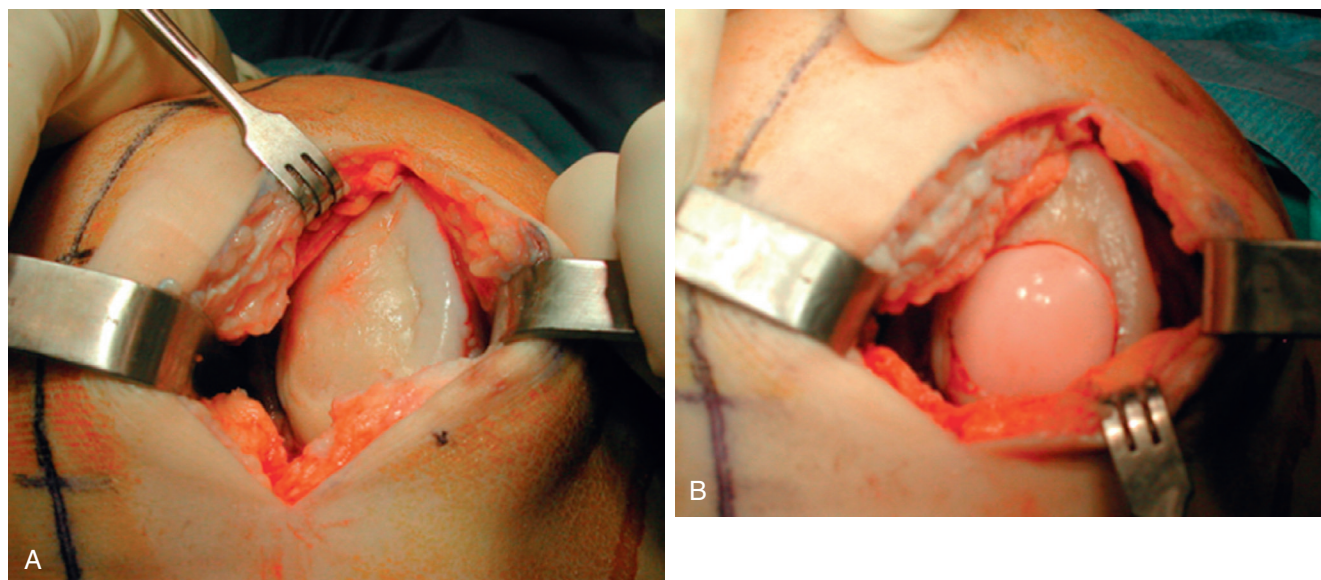


Fig. 97.4 Intraoperative example of (A) a full-thickness focal chondral defect of the medial femoral condyle (right knee) undergoing (B) osteochondral allograft transplantation.

mesenchymal stem cells (hUBC-MSCs), which avoids the donor-site morbidity associated with BMAC harvest. In this technique, the hUBC-MSCs are combined with a sodium hyaluronate scaffold and inserted implanted into the defect site. Preliminary outcomes from a clinical trial out of Korea including seven patients undergoing Cartistem are encouraging, though certainly further research is warranted.⁶⁹

Several other stem cell-matrix combinations for the treatment of focal chondral defects have recently been described, and are currently in the initial phases of clinical testing. Specifically, the implantation of adipose derived stem cells (ADSCs) combined with scaffolds and bioactive factors have been reported by several groups.

Minced Cartilage Techniques

While the DeNovo Natural Tissue (DeNovo NT, Zimmer Biomet, Warsaw, IN) cartilage restoration technique was discussed in the previous chapter, at least two other emerging minced cartilage techniques have been reported, including Cartilage Autograft Implantation System (CAIS) and CartiONE. In brief, minced (or particulated) cartilage restoration techniques involve the placement of either autologous or allogeneic hyaline cartilage into the cartilage defect, combined with a scaffold delivery system, and secured with fibrin glue. The cartilage is minced into 1 to 2 mm³ fragments, which is theorized to allow chondrocytes to escape the extracellular matrix and produce hyaline-type cartilage and, ultimately, integrate with the patient's normal/healthy surrounding cartilage.⁴³⁻⁴⁷

DeNovo NT (see previous chapter for details; Fig. 97.5) is similar to BioCartilage, as both products utilize minced juvenile allograft cartilage, but is different in that the technique does not involve violation of the subchondral bone (no associated microfracture). Cartilage Autograft Implantation System (CAIS; DePuy Mitek, Raynham, MA) is another single-stage minced cartilage technique, but unlike DeNovo NT, CAIS utilizes autologous stem

cells harvested from the patient's intercondylar notch or trochlear border. Following harvest, the cells are minced into 1 to 2 mm³ fragments, combined with a scaffold of polycaprolactone (35%) and polyglycolic acid (65%) with PDO mesh, and fixed to the defect using biodegradable anchors. While preliminary outcomes following the use of CAIS were promising, due to expense and poor patient enrollment, trials involving CAIS in the United States were discontinued.

CartiONE (Orteq Ltd., London, United Kingdom) is another restorative technique involving minced cartilage that can be performed as a single operation. In this technique, autologous cartilage is harvested from the periphery of the patient's trochlea (nonarticulating portion) or intercondylar notch, minced, treated with a patented cell-isolation technology, combined with BMAC, added to a scaffold, and implanted into the defect site. Early clinical outcomes, including histologic outcomes, imaging outcomes, and clinical outcomes, from the INSTRUCT trial utilizing this product have been promising.

Off-the-Shelf Surface Allograft Transplantation

The utilization of osteochondral autografts and allografts was discussed in detail in the previous chapter. Among many reasons, both osteochondral autografts and allografts are advantageous, as they are composed of hyaline cartilage with associated bone and are thus ideal for restoration of articular cartilage defects (especially those with symptomatic bone marrow edema), can be used to treat large defects, and can be used as a revision treatment solution for defects that have failed prior treatment. The main disadvantages of osteochondral autografts include their associated donor-site morbidity and their limited ability to treat large lesions. The main disadvantages of osteochondral allografts include their cost, potentially limited availability (especially outside the United States), and concerns regarding disease transmission. In an effort to provide an osteochondral solution that maintains the benefits but eliminates the disadvantages,

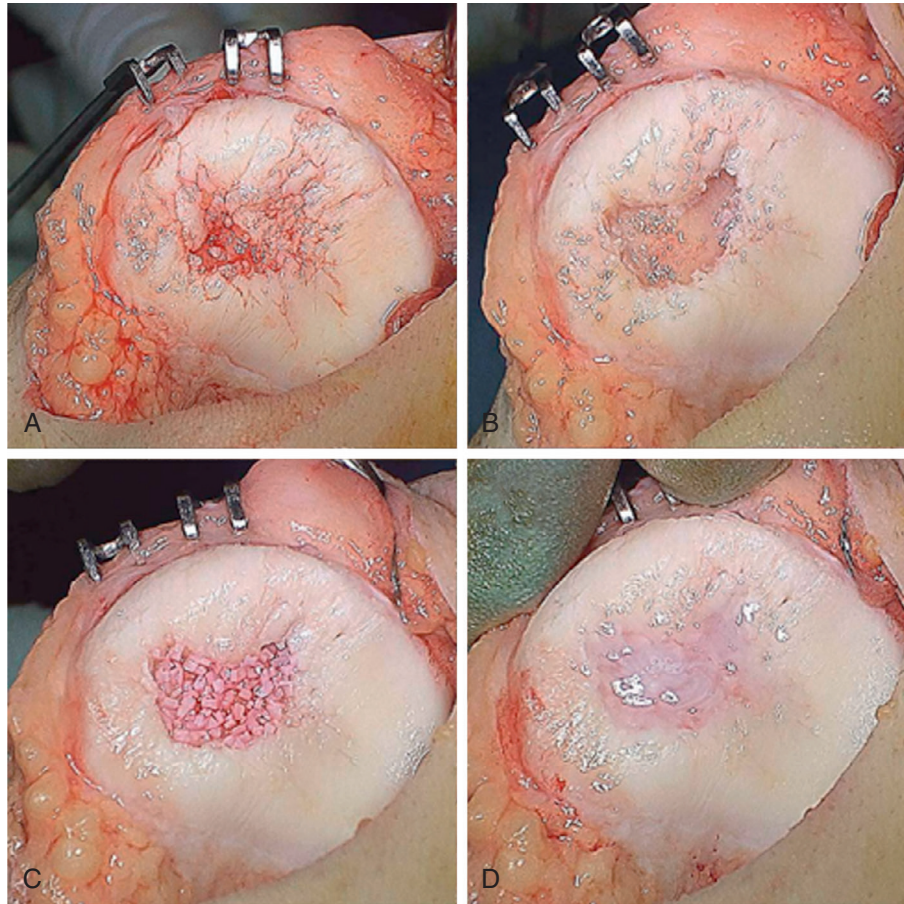


Fig. 97.5 Intraoperative example of a full-thickness focal chondral defect of the patella undergoing DeNovo NT transplantation including (A) defect, (B) preparation of defect with stable vertical walls, (C) transplantation of cells, and (D) covering of defect with fibrin glue.

several off-the-shelf surface allograft products have been developed. Surface allografts available in the United States include Chondrofix (Zimmer Biomet, Warsaw, IN) Cartiform (Arthrex Inc., Naples, FL), and ProChondrix (AlloSource, Denver, CO).

In 2012, Chondrofix was introduced as an off-the-shelf osteochondral allograft for single-stage treatment of full thickness articular cartilage lesions. This product was described as a pre-shaped, cylindrical, sterile, decellularized bone-cartilage construct with a shelf life of 24 months. The graft is precut to 10 mm in length, and comes in either 7, 9, 1, or 15 mm diameters. Preliminary data on 32 patients undergoing treatment with Chondrofix was reported by Farr et al. in 2016, and unfortunately failures were noted in 72% of the cohort (23 knees) at an average follow-up of 1.29 years (range, 0.11 to 2.8 years).^{41,42} The average defect size of the cohort was $2.9 \pm 2.0 \text{ cm}^2$, and a median of 2 allografts were implanted per knee (range, 1 to 5 grafts).

Cartiform and ProChondrix are two more recently described off-the-shelf osteochondral allograft products available in the United States. Unlike Chondrofix, which does not contain any viable chondrocytes, both Cartiform and ProChondrix do contain viable chondrocytes. Cartiform (Fig. 97.6) is described as a cryopreserved, viable osteochondral allograft composed of full-thickness articular cartilage and a thin layer of subchondral bone, with a 24-month shelf-life (stored at -80°C), and

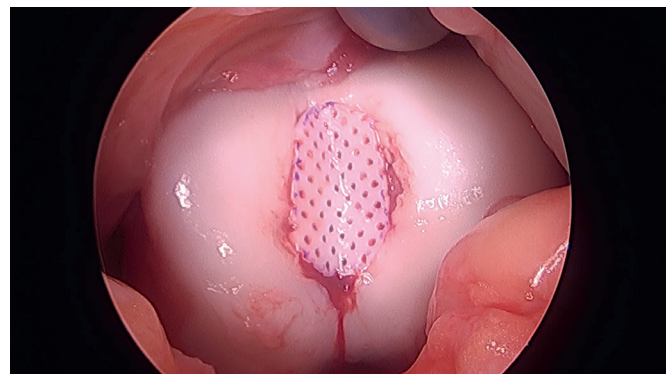


Fig. 97.6 Intraoperative example of a full-thickness focal chondral defect of the trochlea undergoing Cartiform cartilage restoration.

comes in four sizes (10 mm diameter, 20 mm diameter, $12 \times 19 \text{ mm}$ and $20 \times 25 \text{ mm}$). The graft contains full-thickness pores/perforations that allow the cryopreservation solution to bathe the entire graft and preserve cell viability during storage, and also allows for graft flexibility during implantation. ProChondrix is described as a cellular 3D fresh osteochondral allograft composed of viable chondrocytes, matrix, and growth factors, with a 35-day shelf life (stored at 4°C), and comes in 5 sizes

(11, 13, 15, 17, and 20 mm diameter). The depth of the graft can be customized intraoperatively based on the depth of the osteochondral defect. As both of these off-the-shelf osteochondral allograft products are relatively new, clinical outcomes in patients undergoing treatment with these products are currently unavailable.

SUMMARY

Focal chondral lesions of the knee, especially those involving the weight-bearing surfaces of the medial or lateral femoral condyles as well as those involving the patellofemoral joint, often result in pain, effusions, mechanical symptoms, and ultimately dysfunction and disability. For appropriately indicated patients, surgical intervention is helpful in reducing pain, improving function, and restoring normal joint mechanics. Emerging techniques including augmented microfracture, scaffold and matrix-associated constructs, minced cartilage transplantations, MACI, and off-the-shelf osteochondral allograft transplantations, have been described, and warrant additional research and comparison to more “conventional” techniques, including standard chondroplasty, traditional microfracture, ACI, OATS, and OCA.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

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Crawford DC, DeBerardino TM, Williams RJ 3rd. NeoCart, an autologous cartilage tissue implant, compared with microfracture for treatment of distal femoral cartilage lesions: an FDA phase-II prospective, randomized clinical trial after two years. *J Bone Joint Surg Am*. 2012;94(11):979–989.

Level of Evidence:

I

Summary:

Results from a Phase II RCT comparing the safety of autologous cartilage tissue implantation with NeoCart to traditional microfracture surgery.

Citation:

Farr J, et al. High failure rate of a decellularized osteochondral allograft for the treatment of cartilage lesions. *Am J Sports Med*. 2016;44(8):2015–2022.

Level of Evidence:

IV

Summary:

Case series demonstrating a high failure rate following implantation of preshaped, cylindrical sterilized and decellularized osteochondral allografts.

Citation:

Frank RM, et al. Do outcomes of osteochondral allograft transplantation differ based on age and sex? A comparative matched group analysis. *Am J Sports Med*. 2018;46(1):181–191.

Level of Evidence:

III

Summary:

Cohort study demonstrating that osteochondral allograft transplantation is a safe and reliable treatment option for osteochondral defects in patients aged ≥ 40 years. Male and female patients had similar outcomes.

Citation:

Frank RM, et al. The utility of biologics, osteotomy, and cartilage restoration in the knee. *J Am Acad Orthop Surg*. 2018;26(1):e11–e25.

Level of Evidence:

V

Summary:

Review of the latest evidence over the use of biologics, osteotomies, and cartilage restoration.

Citation:

Gille J, et al. Outcome of autologous matrix induced chondrogenesis (AMIC) in cartilage knee surgery: data of the AMIC Registry. *Arch Orthop Trauma Surg*. 2013;133(1):87–93.

Level of Evidence:

IV

Summary:

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Citation:

Moran CJ, et al. Restoration of articular cartilage. *J Bone Joint Surg Am*. 2014;96(4):336–344.

Level of Evidence:

V

Summary:

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